

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461049.9122 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1683 +/- 0.0088
Transit depth (Rp/Rs)^2	2.83 +/- 0.3 [%]
Orbital Inclination (inc)	87.13 +/- 0.89
Airmass coefficient 1 (a1)	1.237 +/- 0.012
Airmass coefficient 2 (a2)	-0.155 +/- 0.089
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	None
Optimal Aperture	3.85
Optimal Annulus	8.7
Transit Duration (day)	0.0506 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1683 $\pm$ 0.0088 vs 0.1594 (deviation 1.01 σ)
Duration fit vs published	0.0506 d vs 0.0512 d (deviation 0.43 σ)
Mid-time O-C	-7.2 min
Depth SNR	19.1
Residual scatter	0.97 %

Light curve

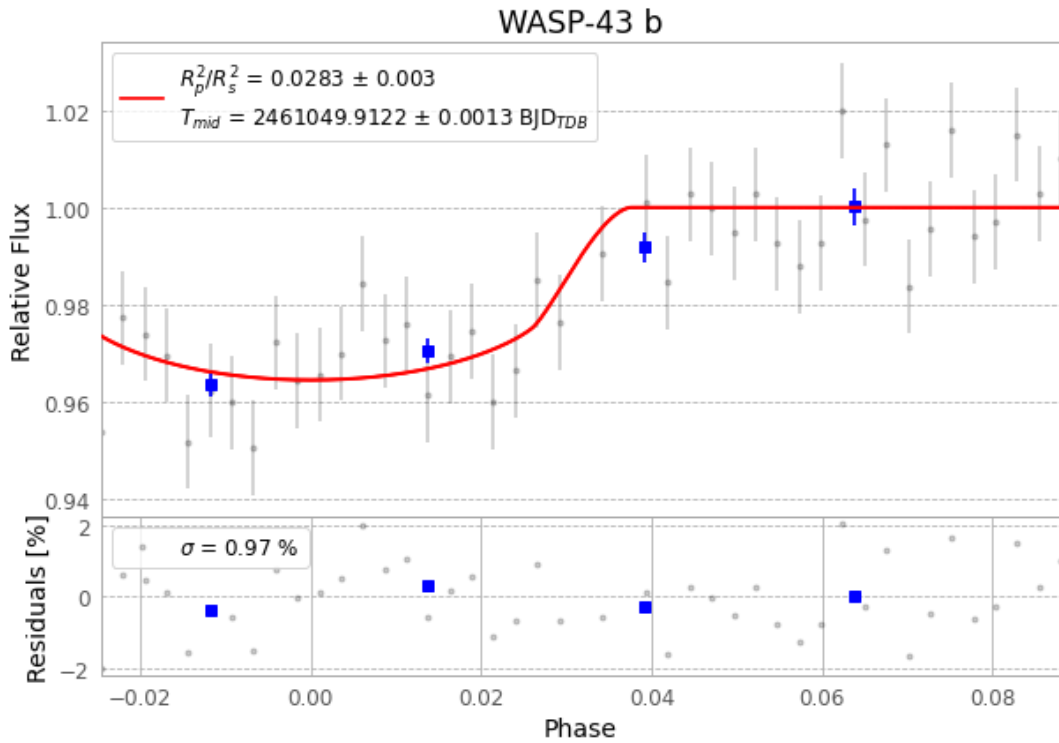


Figure 1 — FinalLightCurve_WASP-43 b_2026-01-09.png (SHA-256 b2214706bae06145...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [425, 321], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2350e6fd7d4cdd4a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

45 FITS frames (30 MB), WASP-43260109091815.FITS → WASP-43260109113016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260109.log (4a8c8f280def...), FinalLightCurve_WASP-43 b_2026-01-09.png (b2214706bae0...), FinalLightCurve_WASP-43 b_2026-01-09.csv (072bcbcca7c6...)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 14:09Z.